Simulating 2d U(1) Lattice Gauge Theory with the Heatbath Algorithm

1 2d Abelian Lattice Gauge Theory

In lattice gauge theory we deal with *link variables* which are elements of the gauge group, U(1) in this case.

If we start from a d-dimensional (hyper-)cubic lattice of size L^d with periodic boundary conditions, there are in total dL^d links, connecting nearest neighbor lattice sites.

So in our case — which is perhaps the simplest non-trivial gauge theory — there are $2L^2$ link variables $U_{x,\mu}$, x being any lattice site and $\mu \in \{1, 2\}$ is the direction of its forward connection to a neighbor site, *i.e.* (in lattice units)

- $U_{x,1}$ connects the site x to $x + \hat{1} = (x_1 + 1, x_2)$
- $U_{x,2}$ connects the site x to $x + \hat{2} = (x_1, x_2 + 1)$.

The backward links are given by $U^\dagger_{x,\mu}$ if the gauge group is SU(N) (Yang-Mills theory), which reduces to $U^*_{x,\mu}$ in the Abelian case,

- $U^*_{x,1}$ connects the site $x + \hat{1} = (x_1 + 1, x_2)$ to x
- $U^*_{x,2}$ connects the site $x + \hat{2} = (x_1, x_2 + 1)$ to x

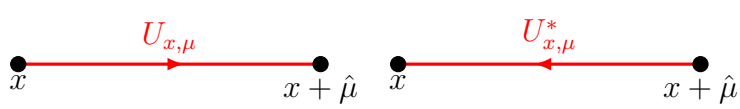

Figure 1: Symbolic illustration of a link variable and its inverse.

If we follow some path on the lattice, we successively multiply the corresponding link variables, for instance (building the product from the right to the left)

$$\ldots U^*_{x+3\,\hat{1}+\hat{2},2} U_{x+2\,\hat{1}+2\,\hat{2},1} U_{x+\hat{1}+2\,\hat{2},1} U_{x+\hat{1}+\hat{2},2} U^*_{x+\hat{1}+\hat{2},1} U_{x+2\,\hat{1},2} U_{x+\hat{1},1} U_{x,1} \ . \quad (1.1)$$

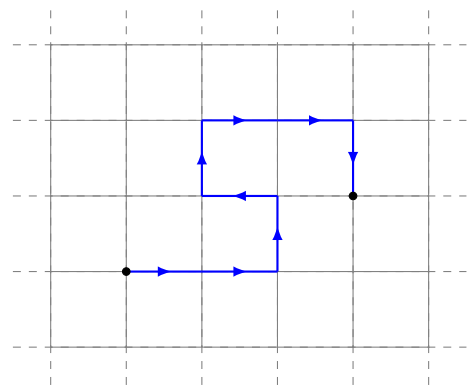

Figure 2: Illustration of the gauge on the lattice, which corresponds to the term (1.1), as an example.

Physical quantities are *gauge invariant*. Here this means the following: we assign to each site x a phase factor $\exp(\mathrm{i}\alpha_x)$ in such a way that

$$U_{x,\mu} = \exp(\mathrm{i}(\alpha_{x+\hat{\mu}} - \alpha_x)) \ , \ \forall x, \ \mu \ . \tag{1.2}$$

Now gauge invariance means that we are free to change the phase factors at the sites, like $\exp(\mathrm{i}\alpha_x) \to \exp(\mathrm{i}\alpha'_x)$ (this is a *local symmetry*). If we implement such a phase change at one site, this affects the link that leads to this site, and the link that emanates from it: the phase changes with the opposite sign. In most cases, in the product for a lattice path these changes cancel. This is not true, however, for a phase change of the first and of the last site of this path, they are marked by black dots in Figure 2, hence such a path is not gauge invariant.

The exceptions are *closed paths*: they are therefore suitable to define gauge invariant quantities, which may have a physical meaning. Such closed paths are known as *Wilson loops*. The simplest (non-trivial) Wilson loop corresponds to the path around one lattice plaquette. It is usually defined in an anti-clockwise orientation,

$$U^{\mathrm{p}}_{x,\mu\nu} := U^*_{x,\nu} U^*_{x+\hat{\nu},\mu} U_{x+\hat{\mu},\nu} U_{x,\mu} \ , \quad (\mu \neq \nu) \ . \tag{1.3}$$

In particular, the plaquette variables are used to construct the Euclidean

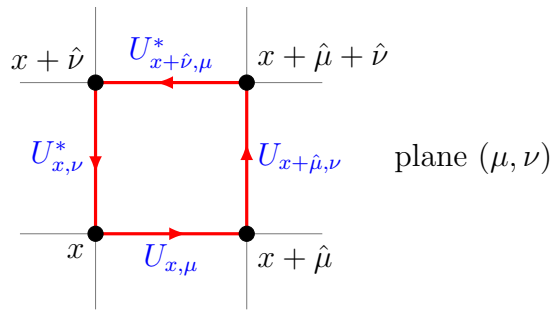

Figure 3: Symbolic illustration of a plaquette variable, which also belongs to the gauge group, U(1) in this case.

action

$$\begin{aligned} S[U] &= \frac{\beta}{2}\sum_{x,\,\mu\neq\nu}(2-U^{\rm p}_{x,\mu\nu}) = \frac{\beta}{2}\sum_{x,\,\mu<\nu}(2-U^{\rm p}_{x,\mu\nu}-U^{{\rm p}\,*}_{x,\mu\nu}) \\ &= \beta\sum_{x,\,\mu<\nu}(1-{\rm Re}\,U^{\rm p}_{x,\mu\nu})\ . \end{aligned} \tag{1.4}$$

For a given volume and value of β (which represents an inverse couplings squared in field theory, or an inverse temperature in Statistical Mechanics), we can perform a simulation and measure *e.g.* the mean plaquette value $\langle U^{\rm p}\rangle \in \mathbb{R}$: for large β (weak coupling) we expect it to approach 1, and as β decreases (strong coupling) it should monotonically increase down towards 0.

2 Heatbath algorithm

Next we discuss the question how to simulate this model. It could be done with the Metropolis algorithm that we are familiar with, by updating the link variable one by one, with the usual Metropolis accept-reject decision.

However, here there is a better option know as the *heatbath algorithm*. That algorithm also updates the link variables one by one, but it ignores its current value and introduces a new phase factor, which is directly obtained based on its probability distribution. This distribution depends on the configuration background, but only as far as it affects a plaquette variable that involves the link to be updated.

Hence the impact of the current configuration is reduced to the "staples" around the link to be updated: a "staple" for the link variable $U_{x,\mu}$ is a lattice path that consists of three links, and represents a next-to-minimal manner to proceed from x to $x+\hat{\mu}$. In d dimensions there are $2(d-1)$ staples to a given link (two in each direction, excluding the direction of the link under consideration).

What ultimately matters is the sum over these staples, which reads

$$\Sigma_{x,\mu} := \sum_{\nu\neq\mu} \left[U^*_{x+\hat{\mu},\nu} U_{x+\hat{\nu},\mu} U_{x,\nu} + U^*_{x-\hat{\nu}+\hat{\mu},\nu} U_{x-\hat{\nu},\mu} U^*_{x-\hat{\nu},\nu} \right] . \tag{2.1}$$

[Side-remark: For Yang-Mills theories, the link variables belong to a gauge group SU(N), $N \geq 2$, and the inverse links are given by Hermitian conjugation instead of just complex conjugation.]

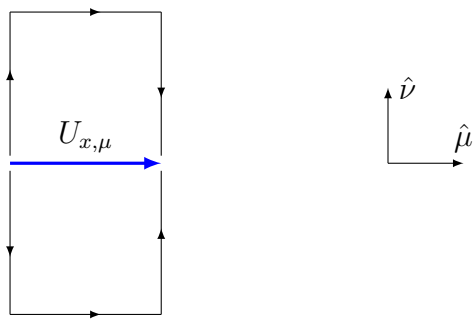

Figure 4: Illustration of the two staple terms in two dimensions, to be summed over, to obtain the term $\Sigma_{x,\mu}$ in eq. (2.1).

In 2d Abelian gauge theory there is just one direction $\nu \neq \mu$, and the Hermitian conjugate simplifies to complex conjugation. The relevant contribution to the action, which is affected by $U_{x,\mu}$, takes the form

$$s[U] = -\beta \mathrm{Re}\left(U_{x,\mu}\Sigma^*_{x,\mu}\right) . \tag{2.2}$$

This is the term that determines the probability distribution that the heat-bath algorithm refers to. If we write $\Sigma_{x,\mu} \in \mathbb{C}$ as $\Sigma_{x,\mu} := se^{\mathrm{i}\gamma}$ (with $s > 0$, $\gamma \in [0, 2\pi)$), the probability density for the updated phase $\varphi'_{x,\mu}$ is given by

$$p(\varphi'_{x,\mu}) = \exp\left(\beta \mathrm{Re}\, s \exp(\mathrm{i}(\varphi'_{x,\mu} - \gamma)\right) = \exp\left(\beta s \cos(\varphi'_{x,\mu} - \gamma)\right) . \tag{2.3}$$

Note that this is not strictly a probability density because it is not normalized, but for the following steps the normalization is not needed.

2.1 Explicit implementation

Although this algorithm does not need a stochastic accept-reject step of the Metropolis type, its simplest implementation involves nevertheless an accept-reject decision, which is, however, deterministic.

From a general perspective, we consider a distribution $\rho(x)$ (which does not need to be normalized), and we denote its maximum as $\rho_{\max}$. Now this method works by drawing a pair of two random numbers (α, b): α is uniform in the full range of possible x-values, and b is uniform as well, but in the interval $[0, \rho_{\max}]$. If $b \leq \rho(a)$ then we accept α as the new variable, otherwise it is rejected and one repeats the procedure until acceptance. In this way, α is indeed distributed with probability density $\propto \rho(x)$, as desired.[1]

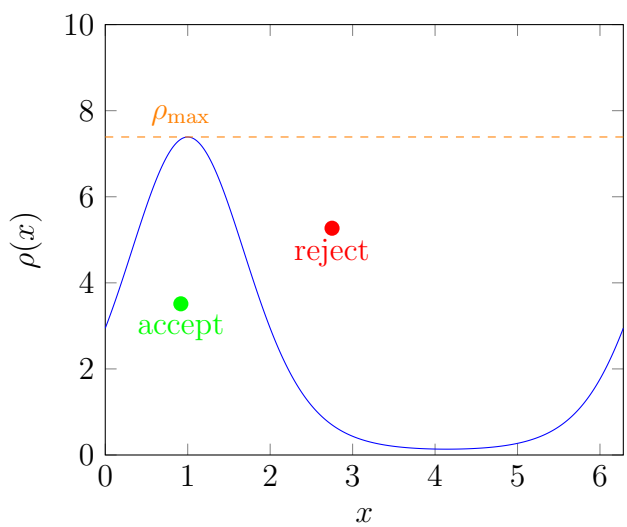

Figure 5: Symbolic illustration of the accept-reject decision in the simplest implementation of the heatbath algorithm.

In the case of 2d lattice U(1) gauge theory, this leads to the following recipe for the update of a link variable $U_{x,\mu} \in \mathbb{C}$, $|U_{x,\mu}| = 1$:

[1] We could transform it into the actual probability density by rescaling, $p(x) = \rho(x)/\int dx\, \rho(x)$, but we do not even need to do so.

1. Compute the staple term $\Sigma_{x,\mu} = s e^{\mathrm{i}\gamma}$, as well as $\rho_{\max} = e^{\beta s}$.

2. Draw a random number α uniformly in $\alpha \in [0, 2\pi)$.

3. Draw another random number b uniformly in $b \in [0, \rho_{\max})$.

4. If
$$b \leq \exp\Big(\beta\sigma\cos(\alpha - \gamma)\Big) \tag{2.4}$$
then accept this suggestion, and set $\varphi'_{x,\mu} = \alpha$. At this point we leave this sequence and proceed to the next link variable.

5. Otherwise return to step 2 and repeat the procedure until some variable α is accepted, and turned into $\varphi'_{x,\mu}$.

The simulations runs as usual: here a cold start means that we initially set all link variables to 1, and for a hot start we choose for each link an independent random phase, uniformly in the interval $[0, 2\pi)$.

Then we have to perform a sufficient number of thermalization sweeps, and take measurements which are separated by some sweeps in the Markov chain.

2.2 Observables

- The simplest observable is the mean plaquette value, which is directly related to the energy density
$$\epsilon = \langle \mathrm{Re}\, U^{\mathrm{p}} \rangle \ . \tag{2.5}$$
Here the average refers to the set of configurations, as well as the sites x and (in $d > 2$) the possible plaquette orientations. Results as a function of β, in a lattice volume 100×100, are shown in Figure 6.

- The topological charge density is given by the complex phase of a plaquette. For simplicity we write it down in two dimensions,
$$q_x = \arg U^{\mathrm{p}}_{x,12} \in (-\pi, \pi] \ , \quad Q = \frac{1}{2\pi}\sum_x q_x \in \mathbb{Z} \ . \tag{2.6}$$
The property of the total topological charge Q of a configuration to be integer-valued is due to the periodic boundary conditions.

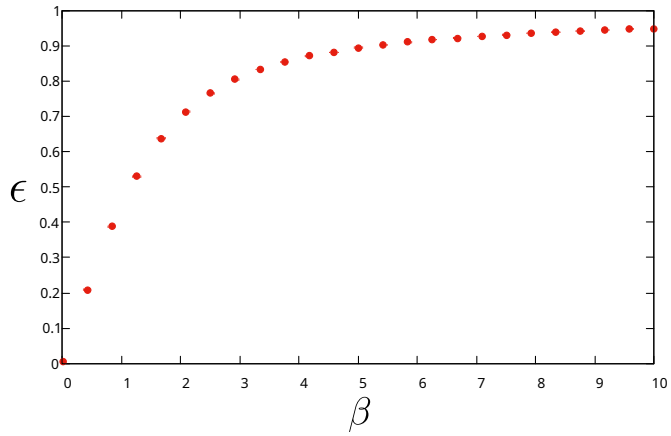

Figure 6: Energy density ϵ as a function of β, obtained by simulations on a 100×100 lattice.

As a consistency check, one should first verify that $\langle Q \rangle$ is compatible with 0 (no sign of q_x is favored).

Next we can measure the topological susceptibility,

$$\chi_{\rm t} = \frac{\langle Q^2 \rangle - \langle Q \rangle^2}{V} \ , \tag{2.7}$$

where V is the lattice volume. In practice one may or may not subtract in the numerator the term $\langle Q \rangle^2$, which theoretically vanishes due to parity symmetry. As in the chapter about the 1d XY model, we could proceed with the kurtosis and its ratio with $\chi_{\rm t}$ (and higher Binder cumulants, if you wish).

- Finally we proceed to the *Polyakov loop*. It is a specific Wilson loop: it is closed thanks to the periodic boundaries in the Euclidean time direction, which we denote as x_2. Let us assume here a rectangular lattice of size $L_1 \times L_2$. For some coordinate x_1, the Polyakov loop is given by

$$P_{x_1} := U_{(x_1,L_2),2} \ldots U_{(x_1,3),2} U_{(x_1,2),2} U_{(x_1,1),2} = \prod_{x_2=0}^{L_2-1} U_{(x_1,L_2-x_2),2} \ , \tag{2.8}$$

i.e. we multiply the link variables following a path over the entire volume in the 2-direction. Two such Polyakov loops are sketched in Figure 7.

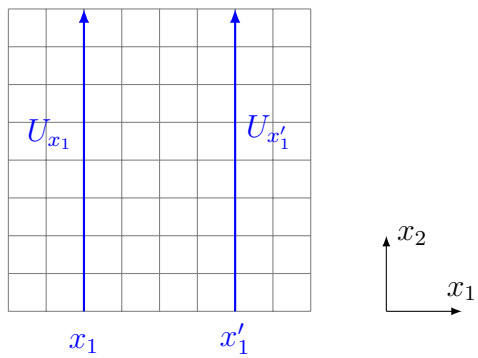

Figure 7: Sketch of two Polyakov loops, P_{x_1} and $P_{x_1'}$.

The product $P_{x_1} P^*_{x_1'}$ can be interpreted as the energy which is required for the creation of a pair of particles with opposite charge (the charge is the coupling to the gauge field). These particles — we might call them an "electron" and a "positron" — are static at x_1 and x_1', respectively, and they exist during the (Euclidean) time L_2. The expectation value of this product of Polyakov loops determines the potential $V(r)$ between these static particles, with $r := |x_1 - x_1'|$, according to the proportionality relation

$$\langle P_{x_1} P^*_{x_1+r} \rangle \propto \exp(-L_2 V(r)) \ , \tag{2.9}$$

at least in the large-L_2 limit. The final goal of this project is the identification of this potential $V(r)$.

In general, for instance when referring to static "quarks" in SU(3) gauge theory, this potential is parameterized as

$$V(r) = A + \sigma r + \frac{B}{r} \ , \tag{2.10}$$

where A, σ, B are constants. The term $\propto 1/r$ dominates at short distance r: in QCD it is known as the Lüscher term, and it can be derived from the Casimir effect. In our case, however, we only expect one term $V(r) \propto r$, and its coefficient σ is known as the *string tension*. The picture is that one imagines some "string" to bind the particle-antiparticle pair, and to follow Hook's law, which implies confinement.